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THE RELEVANCE OF INFORMATION THEORY TO MODERN CRYPTOGRAPHY

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ABSTRACT

The information-theoretic and the complexity-theoretic approaches to cryptography are contrasted, as are also the notions of an unconditionally secure secrecy system and a computationally secure secrecy system. The reasons for a certain pessimism at the present time about both approaches are identified. It is argued that although the main successes in cryptography in the recent past have come about through the complexity-theoretic approach, the information-theoretic approach may have the better chance of leading to provably-secure practical ciphers. A recently developed information-theoretic approach, due to Maurer, is described and is used to illustrate the power and versatility of information-theoretic techniques in cryptography.

INTRODUCTION

Although one can argue quite fairly that the first truly scientific paper on cryptography was Shannon's 1949 paper [1] whose centerpiece was the formulation of the information-theoretic approach to cryptography, one must also in fairness concede that information theory has played only a minor role in the revolutionary developments within cryptography over the past fifteen years. The intellectually most exciting new ideas and the practically most promising new techniques that have emerged in this period, such as public key-distribution systems, public-key cryptosystems, zero-knowledge proofs, and a vast array of baffling cryptographic protocols, all owe their existence exclusively to the complexity-theoretic approach to cryptography. One could not fault an observer of these developments who concluded that the information-theoretic approach, after an impressive start, has simply petered out.

Has information theory really lost its relevance to cryptography? To answer this question, as we shall attempt to do in the sequel, requires that we have a clear understanding of the methodologies of information theory and of complexity theory with respect to their

applications in cryptography. To develop this understanding, we shall confine ourselves to the consideration of secrecy systems, but we remind the reader that the twin goals of cryptography, namely secrecy and authenticity, are of equal importance and interest.

UNCONDITIONAL SECURITY vs. COMPUTATIONAL SECURITY

What Shannon referred to in 1949 as "theoretical secrecy" and "practical secrecy" are today more commonly called "unconditionally secure secrecy" and "computationally secure secrecy", respectively, but Shannon's characterizations of these concepts are as fresh as ever. A secrecy system is *unconditionally secure* if it is "immune ... to cryptanalysis when the cryptanalyst has unlimited time and manpower available" [1, p.679]. A secrecy system is only *computationally secure* if "breaking it is equivalent to (or requires at some point in the process) the solution of some problem known to be laborious" [1, p. 704]. This dichotomy of security seems at first glance to be in a one-to-one correspondence with the dichotomy between information theory and complexity theory, and indeed this is the common belief of cryptographers today: information theory deals with unconditional security whereas complexity theory deals with computational security. One of our main tasks is to call this correspondence into question, but first we give an example of a cryptosystem that illustrates the concept of unconditional security.

Example 1:

Consider the secrecy system of Fig. 1 in which the *plaintext* X and the *secret key* Z are independent random variables taking values in the finite group G , whose group operation we denote by $\cdot$. The secret key Z is chosen so as to be equally likely to be any of the elements of G . The *ciphertext* Y is formed as $Y = X \cdot Z$. The intended recipient, who also knows the secret key Z and thus can compute its inverse Z' in the group, can recover the plaintext as $Y \cdot Z' = X \cdot Z \cdot Z' = X$.

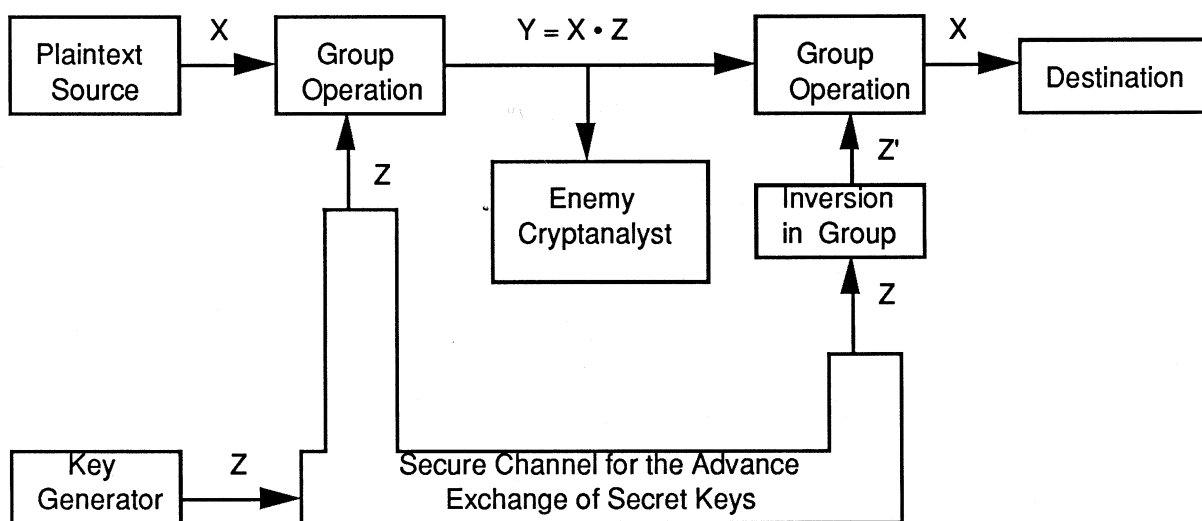


Fig. 1: Example of a secrecy system that is unconditionally secure.

The enemy cryptanalyst, who is presumed to know the statistics of X and Z , observes only the cryptogram Y (i.e., a *ciphertext-only attack* is assumed). Because $P(Y=y|X=x) = P(x \bullet Z = y|X=x) = P(Z = x' \bullet y|X=x) = P(Z = x' \bullet y) = 1/\#(G)$ for all x and y in G , where $\#(G)$ denotes the number of elements in G , it follows that the cryptogram Y is statistically independent of the plaintext X or, equivalently in information-theoretic terms, the mutual information $I(X;Y)$ is 0. It follows that no amount of computation by the enemy cryptanalyst can be of any avail in determining any information whatsoever about the plaintext X . Thus, this secrecy system provides unconditional security against a ciphertext-only attack and, moreover, the afforded secrecy is total. Shannon referred to systems of this type as providing *perfect secrecy*. (Note that the conclusion that this system provides perfect secrecy holds regardless of the statistics for the plaintext X .)

The reader will recognize the system of Fig. 1 as a slight generalization of Vernam's cipher [2] or the *one-time pad* as it is more commonly called today. In Vernam's cipher, G is the abelian group of all n -tuples with components in the L -ary alphabet $\{0, 1, \dots, L-1\}$ and the group operation $\bullet$ is component-wise addition modulo L . The "one-time" terminology arises from the fact that the unconditional security of the system depends strongly on the fact that the system is assumed to be used only one time. If a sequence of plaintexts are to be sent, then to retain unconditional security the secret key must be chosen anew for each such plaintext, i.e., each key may be used only one-time. Our reasons for making this slight generalization of Vernam's cipher are that we have found this perfect-secrecy property of group transformations to be very useful in designing practical ciphers [3] and that essentially the same property lies at the heart of many zero-knowledge-proof protocols.

SHANNON'S PESSIMISTIC INEQUALITY

Perhaps the highlight of Shannon's information-theoretic treatment of secrecy in [1] was his proof that, for a secrecy system to provide perfect secrecy (i.e., to be unconditionally secure and to offer total secrecy), then the *uncertainties* (or *entropies*) of the secret key Z and the plaintext X must satisfy

$$H(Z) \geq H(X). \quad (1)$$

The simple information-theoretic proof goes as follows: Because $I(X;Y) = H(X) - H(X|Y) = 0$ by the hypothesis of perfect secrecy, it follows that $H(X) = H(X|Y) \leq H(ZX|Y) = H(Z|Y) + H(X|YZ) = H(Z|Y) \leq H(Z)$, where we have used the fact that $H(X|YZ) = 0$, which is just the condition that the intended recipient should be able to recover the plaintext X from the cryptogram Y and the secret key Z .

The conclusion to be drawn from Shannon's inequality (1) is a very pessimistic one indeed. For this inequality tells us that if we desire our secrecy system to be unconditionally secure and to offer total secrecy, then we cannot avoid using as many binary digits of randomly chosen secret key as there are bits of information in the plaintext to be safeguarded. The practical problems of generating and securely exchanging so much secret key prohibits the cryptographer from even considering such systems for all but the most exotic applications. Our own experience suggests that

it is precisely this inequality (1), whose proof is so ironclad and whose content is so discouraging, that has dampened the enthusiasm of cryptographers for information theory--and perhaps caused them to have overlooked its usefulness.

The reader should have no difficulty (cf. [4]) showing that inequality (1) still applies for perfect secrecy when the encrypting process is randomized, i.e., when the cryptogram Y is determined not only by the plaintext X and the secret key Z but also by a *randomizer* R (which implies that $H(Y|XZR) = 0$), where R is known only by the sender. The condition $H(X|YZ) = 0$ is thus still the condition for the intended recipient to be able to recover the plaintext. Shannon never considered such a randomizer in his information-theoretic treatment of secrecy [1], perhaps because he knew that the pessimistic inequality (1) could not thereby be avoided. We shall see, however, that randomizers of a somewhat different sort can be very useful in extending the information-theoretic approach to secrecy.

PESSIMISM AMID COMPLEXITY

In spite of its many important theoretical and practical contributions to cryptography over the past fifteen years, the complexity-theoretic approach has also been the object of increasing pessimism among cryptographers. There are two reasons for this. First, the computational security of the cryptographic techniques that have arisen from complexity theory are based on only a small number of "difficult problems" (taking discrete logs, factoring the product of large primes, and solving a 0-1 knapsack--cf. [5]), which is a very small base for such a large number of techniques, and, second, none of these problems has been proved to be a difficult one. (The reader may object to this last assertion on the grounds that the knapsack problem, at least, is known to be NP-complete and thus is certainly a "difficult problem", but he should remember that the supposed difficulty of NP-complete problems rests on the conjecture that NP is not equal to P.) The hard fact is that, to date, complexity theory has yielded only the following kind of "if-then" results in cryptography: *If* this problem is difficult, *then* this secrecy system is computationally secure against the specified attack. More worrisome is the fact that it is an open question whether it is even possible in principle to design a secrecy system whose breaking is "equivalent to (or requires at some point in the process) the solution of some problem known to be laborious"--as opposed to presumed to be laborious. It could be that complexity theory is eternally doomed to the production of "if-then" results in cryptography--and it could be that tomorrow some complexity theorist will produce a practical cipher whose breaking is provably equivalent to solving a provably hard problem.

MAURER'S INFORMATION-THEORETIC APPROACH

Very recently, Maurer [6],[7] has suggested a new twist to the information-theoretic approach to secrecy that avoids the roadblock of Shannon's inequality (1). Maurer's innovation was to introduce an event S , which we shall call the *security event*, that has the property that if

this event occurs then the secrecy system is unconditionally secure. But if the event S does not occur, the system might be very insecure. Shannon's perfect secrecy can in this light be interpreted as the special case where S is the certain event and where the afforded secrecy is total. As we shall see in an example, it is possible in Maurer's formulation simultaneously to have total and unconditionally secure secrecy when the event S occurs, to have S an event of negligibly small probability, and still to have $H(Z) \ll H(X)$ --i.e., to violate strongly the pessimistic inequality (1)! Maurer arrived at his formulation in the process of analyzing the "provably-secure but totally impractical" Rip van Winkle cipher [8], but we will illustrate the approach with a simpler and earlier cipher that is also provably-secure (and even less practical). This cipher is due to Diffie in unpublished work that he called to our attention [9] upon hearing about the results in [8].

Example 2:

Diffie's secrecy system exploits the existence of a very large number, 2^K , of telephones that can be dialed by anyone. Note that a K bit telephone number suffices to identify each of these telephones. The i -th telephone, when dialed, plays back a recorded binary sequence R_i of length N (where $N \gg K$) that was obtained exclusively for that telephone by coin-tossing. The secret key Z is a K -bit telephone number that is equally likely to be that of any of the 2^K telephones. The secret key is known to both the sender and intended recipient, but not of course to the enemy cryptanalyst. The system works as follows: When he wishes to send an N -bit plaintext X to the intended recipient, the sender dials the telephone number Z and obtains the random sequence R_Z . The sender then adds R_Z to X using component-wise modulo-two addition (just as in the $L=2$ Vernam cipher) to produce the N bit cryptogram Y that he then sends over an insecure channel to the intended recipient. The recipient, upon receipt of Y , also dials the telephone number Z , obtains the same random sequence R_Z , and subtracts this from Y component-wise modulo-two (which is the same as addition) to obtain the plaintext X .

The security event S is the event that the enemy cryptanalyst, who intercepts the message Y and then begins his attack to find X , never dials the telephone number Z in the course of his attack. Unless he dials Z , the sequence R_Z will be a random binary sequence that is statistically independent of all the other data that this cryptanalyst has obtained, i. e., it will be equivalent to the secret key in an $L=2$ Vernam cipher. Thus, given that the event S occurs, Diffie's system provides total secrecy and is unconditionally secure. By a trivial calculation, if the enemy cryptanalyst dials t different telephone numbers in the course of his attack, then the probability $P(S^c)$ that the security event S does not occur is $t/2^K$. For instance, if the secret key is only $K = 100$ bits long, then an enemy cryptanalyst who dialed $t = 2^{50} \approx 10^{15}$ different telephones during his attack would have probability only $P(S^c) = 2^{50}/2^{100} \approx 10^{-15}$ of obtaining any information about the plaintext. But N can be arbitrarily larger than K or, equivalently, $H(X)$ can be arbitrarily larger than $H(Z)$ so that Shannon's inequality (1) can be violated by as much as one wishes!

The only practical problem with the special case of Diffie's system just described would be getting someone to install the required $2^{100} \approx 10^{30}$ telephones, each loaded with its own random binary sequence. One should note, however, that any number of sender/intended-recipient pairs could use the system once it was installed—it might well be a cost-effective secrecy system for a planet with about 10^{30} inhabitants!

Maurer [8] has developed secrecy systems based on his newly formulated information-theoretic approach that offer unconditionally secure and total secrecy when the security event S occurs, that force the cryptanalyst to do infeasibly much work to make the probability that S does not occur non-negligible, and that, if not quite practical for this planet, are at least on the edge of being so.

CONCLUDING REMARKS

The reader has by now foreseen our answer to the question: Is information theory irrelevant to modern cryptography? We would answer this question with a resounding NO! It is not that information theory is irrelevant, but rather that we information theorists working in cryptography have too often joined in the pessimism about our discipline's potential for contributions in this field instead of seeking to make the extensions to Shannon's theory [1] needed for its effective application in cryptography.

Maurer's new information-theoretic approach to secrecy strikes us as the kind of clever twist that we need to add to Shannon's theory if we are ever to reach every cryptographer's dream, the provably-secure and practical cipher. It is interesting to note that both in block-to-block source coding and in most forms of channel coding, very little interesting theory can be developed if one demands absolutely zero error probability at the destination. It is only by allowing a *positive*, albeit arbitrarily small, error probability that one arrives at Shannon's beautiful characterization of a source by its rate and of a channel by its capacity. Can this also be the reason that, in secrecy coding, Maurer's allowance of a small positive probability of insecurity seems to open the door to a practical information-theoretic approach to secrecy? It is hard to believe that Maurer's approach is the only way to open this door that has too long been barred by Shannon's inequality (1) and we encourage other information theorists to exercise their ingenuity in finding them.

We feel compelled to add that there is an even more urgent need for inspired information-theoretic research in authentication techniques. This aspect of cryptography has only in the past decade received, in the work of Simmons (cf. [4] for a review thereof), an information-theoretic basis reasonably akin to that for secrecy given by Shannon. This is partly because authenticity is a much more elusive subject than secrecy, and partly because the potential users of cryptography have only recently become aware of the enormous practical importance of authenticity.

To end this paper on a very honest note, we must say that we have as yet seen no way based on information theory to formulate zero-knowledge proofs and similar protocols—which is

of course not to say that such ways do not exist; we hope the reader will take our lack of success as a challenge to find such ways.

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